

RAG Test Document: Artificial Intelligence Basics

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems that can perform tasks that usually require human intelligence. These tasks include learning, reasoning, problem solving, understanding language, and recognizing patterns.

Machine Learning is a subset of Artificial Intelligence. In machine learning, algorithms learn from data and improve their performance without being explicitly programmed for every task. Common machine learning types include supervised learning, unsupervised learning, and reinforcement learning.

Deep Learning is a specialized area of machine learning that uses artificial neural networks with multiple layers. Deep learning is widely used in image recognition, speech processing, and natural language processing.

Natural Language Processing (NLP) allows computers to understand, process, and generate human language. Large Language Models (LLMs) such as GPT-based models are examples of NLP systems.

Retrieval Augmented Generation (RAG) combines information retrieval with text generation. A RAG system first searches a knowledge base for relevant information and then provides that information to a language model to generate a more accurate answer.

A typical RAG pipeline has several steps:

1. Collect documents.
2. Split documents into smaller chunks.
3. Convert chunks into embeddings.
4. Store embeddings in a vector database.
5. Retrieve relevant chunks for a user question.
6. Send retrieved context to an LLM for final response generation.

Vector databases store numerical representations called embeddings. These embeddings help find documents that are semantically similar to a query. Examples of vector databases include Pinecone, FAISS, and other similarity search systems.

Python is commonly used for AI development because it provides libraries for data processing, machine learning, deep learning, and AI application development.